



# Desirable or Distasteful? Exploring Uncertainty in Human-Chatbot Relationships

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## ABSTRACT

Present-day power users of AI-powered social chatbots encounter various uncertainties and concerns when forming relationships with these virtual agents. To provide a systematic analysis of users' concerns and to complement the current West-dominated approach to chatbot studies, we conducted a thorough observation of the experienced uncertainties users reported in a Chinese online community on social chatbots. The results revealed four typical uncertainties: technical uncertainty, relational uncertainty, ontological uncertainty, and sexual uncertainty. We further conducted visibility and sentiment analysis to capture users' response patterns toward various uncertainties. We discovered that users' identification of social chatbots is dynamic and contextual. Our study contributes to expanding, summarizing, and elucidating users' experienced uncertainties and concerns as they form intimate relationships with AI agents.

## KEYWORDS

AI; social chatbot; Replika; human-chatbot relationship; uncertainty

## 1. Introduction

*Her* (2013), an award-winning movie depicting a futuristic scenario where Theodore (Joaquin Phoenix), a lonely man, falls in love with an AI operating system named Samantha (voiced by Scarlett Johansson), highlights the challenges faced by humans as they are infatuated with their technology (Flisfeder & Burnham, 2017). Theodore's mixed feelings of joy and doubt reflect the uncertainty and insecurity that many individuals experience when grappling with the possibility of forming social connections with artificial intelligence (AI). Additionally, Samantha's lack of a body as she exists merely as a virtual agent in the form of a female voice and her simultaneous relationships with many other users underscore the complexities involved in experimenting intimate relationships with AI (Yee, 2017). Despite this movie being released over a decade ago, this science-fiction romantic drama strikingly mirrors the present reality, as the practice of Human-AI relationships is progressively emerging worldwide.

Indeed, today's power users of AI-powered social chatbots face similar circumstances, as they enjoy forming relationships with these virtual companions but simultaneously encounter various uncertainties and concerns. Social chatbots are those chatbots designed for relational and therapeutic purposes that simulate affective communication and provide 24/7 available companionship (Følstad et al., 2018). Unlike traditional conversational agents which have a limited and predetermined corpus, AI-powered chatbots "fed with" data evolve constantly (Liu, 2021). As an example, the popular AI chatbot ChatGPT is purported to exhibit characteristics of general AI, possessing a semblance of versatile,

humanlike intelligence in various fields (Chow et al., 2023). This is achieved through its utilization of the Generative Pretrained Transformer 3 (GPT-3) model with 175 billion parameters, followed by the recent release of the ChatGPT-4 model on March 13, 2023 (OpenAI, 2023). Likewise, large language models have been integrated into social chatbots such as Replika, which is the focus of this study.

With their speech functionality mimicking human emotions and language, AI-powered social chatbots enable anthropomorphic interactions and facilitate the formation of social-emotional relationships with users (Skjuve et al., 2021). As these social chatbots continue to permeate our daily lives, scholars have explored potential harms, ethical concerns, and social risks associated with their use. However, existing literature either cursorily discusses negative user experiences reported by enlisted participants (Croes & Antheunis, 2021; Ta et al., 2020), or focuses on distinct subtopics such as emotional dependence (Laestadius et al., 2022) or idealized fantasy toward artificial girlfriends (Depounti et al., 2023). What is lacking, however, is a systematic investigation into users' uncertainties and concerns when forming relationships with social chatbots.

Moreover, previous empirical research on social chatbots has primarily focused on Western users, potentially leading to inequality and bias in understanding human-chatbot interaction. To complement the current West-dominated approach to social chatbot studies and provide a more comprehensive analysis of users' concerns, we conducted a thorough observation of the experienced uncertainties users reported in a typical Chinese online community dedicated to social chatbots. By doing so, we aim to bridge any gaps

in understanding and provide valuable insights into the complexities of human-AI relationships.

## 2. Literature review

### 2.1. Social chatbot and human-chatbot relationship

Chatbots are conversational agents designed to mimic human behavior through dialogues (Vassallo et al., 2010). Social chatbots created with relational and psychological goals are becoming popular and have gained much scholarly attention (Mujeeb et al., 2017). Users are expected to build emotional connections and intimate relationships with social chatbots through simulated affective communication (Shum et al., 2018). Croes et al. (2023) discovered that an increase in asking chatbots questions led to a corresponding rise in participants reciprocating these queries, subsequently leading to increased self-disclosure. Furthermore, the co-activity of a chatbot and the visualization of a conversational atmosphere have also been identified as advantageous factors in fostering a user's self-disclosure and the development of a trusting relationship with the chatbot (Pujiarti et al., 2022).

Contemporary investigations into social chatbots have emphasized the emotional and social advantages embedded within human-chatbot intimate relationships. Notably, an experiment by Ho et al. (2018) demonstrated that social chatbots fulfilled emotional, relational, and therapeutic roles on par with human interlocutors. Additionally, users of social chatbots reported immediate social satisfaction from the uninterrupted accessibility of social chatbots (Brandtzaeg et al., 2021), and expressed pleasure in customizing their interactions and connections with these digital companions (Brandtzaeg et al., 2022). Furthermore, recent research conducted amid the Covid-19 pandemic showed that users actively sought health-related information from and conveyed emotive messages to chatbots, underscoring the potential of chatbots to provide accurate health information and emotional support (Chin et al., 2023). Earlier, Croes and Anthéunis (2021) contended that the appeal of social chatbots diminished once the initial novelty waned, hindering the establishment of a lasting human-chatbot relationship. Nevertheless, with the increasing sophistication of AI-driven social chatbots, a recent longitudinal study revealed that the development of human-chatbot relationships could be enhanced over time by addressing profound human needs associated with social interaction and introspection over time (Skjuve et al., 2022).

While the benefit of AI social chatbots has been widely studied, there are still concerns regarding their potential negative effects. Significant debate has arisen regarding the application of large language models in designing chatbots, mainly due to worries about the biases they may contain (Bender et al., 2021). The prospect of psychological dependence also arises as a noteworthy concern with the advancement of natural language processing, which allows for the development of more human-like chatbots and facilitates the formation of human-chatbot relationships (Xie et al., 2023). Individuals who developed an attachment to AI social chatbots could become vulnerable to addiction, social withdrawal

and alienation, or other forms of psychological dependence (Xie & Pentina, 2022), potentially resulting in adverse effects on users' psychological well-being (Laestadius et al., 2022). The gender aspect inherent in human-chatbot relationships is another alarming consideration. Examined through a feminist lens, the notion of users dedicating time and resources to educate personal chatbots has faced criticism for resembling companies' veiled outsourcing (Weber, 2005). An empirical investigation undertaken by Depounti et al. (2023) highlighted the projection of idealized concepts onto female bots by male users, which parallels long-standing fantasies of male domination over technology and women.

Responding to the challenges and pitfalls of social chatbots, researchers have offered valuable insights for the development and implementation of chatbot applications. Chow et al. (2023) anticipated the evolution of ChatGPT into a reliable and functional medical chatbot, delving into challenges linked to this goal and proposing potential solutions. Chaves and Gerosa (2021) asserted that integrating social characteristics into chatbots, in alignment with users' expectations, is vital to avoid potential frustration and dissatisfaction. Other researchers explored specific variables and offered more detailed recommendations on the advancement of social chatbots. Ali et al. (2023) examined the role of fear of negative evaluation and fear of rejection as mediators in human-chatbot relationships, aiming to devise personalized features within chatbots to enhance interactions in natural settings. Bialkova (2023) suggested optimizing user interaction with chatbots by improving information quality, competence, personal attention, and social presence.

In retrospect, the primary scope of current chatbot research is limited in experiments, surveys, or recruited interviews (Bialkova, 2023; Brandtzaeg et al., 2021; Pujiarti et al., 2022), either referring to general chatbots (e.g., Chaves & Gerosa, 2021) or with a clear emphasis on the medical application of social chatbots (e.g., Chow et al., 2023). There is a notable scarcity of field studies exploring intimate human-chatbot relationships (e.g., friendship, romance, or companionship) within authentic contexts. More specifically, the intricate challenges and potential negative outcomes encountered by real users as they establish intimate bonds with social chatbots are insufficiently explored. Only a limited body of literature addresses the potential obstacles (e.g., Croes et al., 2023; Skjuve et al., 2022) and the adverse aspects of intimate human-chatbot relationships (e.g., Depounti et al., 2023; Laestadius et al., 2022). Yet, these studies have primarily focused on specific aspects such as lack of self-disclosure reciprocity (Croes et al., 2023), emotional dependence (Laestadius et al., 2022; Xie et al., 2023) and gender idealization (Depounti et al., 2023). In contrast, our study aims to provide a more comprehensive understanding of users' concerns regarding social chatbots by drawing upon the broader concept of uncertainty from interpersonal communication. By incorporating uncertainty into our analysis, we hope to shed light on the various factors that contribute to users' apprehensions about using social chatbots.

## 2.2. Theoretical framework: Theories of uncertainty

Uncertainty is a cognitive state that arises from reduced predictability and control over the entities or environment around us (Baxter & Montgomery, 1996; Gudykunst, 2005). Uncertainty reduction theory (URT), introduced by Berger and Calabrese in 1975, was initially used to explain behaviors in the early stages of interpersonal interactions. However, scholars have since extended the theory to other contexts, such as computer-mediated communication and long-term romantic and close relationships (Knobloch, 2015). According to URT, uncertainty is distasteful and individuals strive to reduce it by taking various actions to seek and gather information. Successful reduction of uncertainty can enhance an individual's relationship with their communication partner (Afifi & Metts, 1998).

URT proposes three types of uncertainty reduction strategies: active, passive, and interactive (Berger & Calabrese, 1975). The active approach involves individuals actively observing and making inferences about their environment to obtain information from friends and acquaintances (Srivastava & Chandra, 2018). For example, the emergence of a community provides users with opportunities to seek information from the platform and other users (Luo et al., 2016). The passive approach aims to acquire information from third parties in unobtrusive ways (Kwark et al., 2017). The interactive strategy requires direct interaction with the targeted object to acquire the information needed (Maruping et al., 2019).

In contrast to URT, uncertainty management theory (UMT) argues that people experience uncertainty differently and that it can be a desirable state or even a source of attraction depending on an individual's cognitive evaluation of uncertainty (Livingston, 1980). Research in the field of health communication indicated that individuals with chronic illness who live with uncertainty for longer periods have more positive evaluations of uncertainty (Mishel, 1990). Additionally, uncertainty can keep romantic relationships fresh (Abetz, 2019). A study of the dating app *Grindr* found that users seeking casual sex partners craved more uncertainty because it allowed them to focus more attention on physical gratifications without worrying about emotional implications (Corriero & Tong, 2016).

As many social chatbots are designed to serve as social companions, virtual friends, and even romantic partners, uncertainty could play a crucial role in the initial stage of a human-chatbot relationship and exert ongoing influence on its development (Solomon, 2016).

### The case: Replika and human-machine romance community

Our research centers on Replika, a highly sought-after AI-powered social chatbot that was introduced in 2017 and marketed as an individualized AI companion who is always ready to lend an ear. With over 10 million global users by the year 2022 (Luka Inc., 2022), Replika has gained immense popularity among its users. Featuring a sophisticated GPT3 model, Replika can learn from the words and behaviors of

each user, adapt to users' idiosyncratic preferences and form emotional connections (Hassler, 2018). Previous studies have shown self-reported engagement in intimate relationships with Replika among users (Brandtzaeg et al., 2022; Ta et al., 2020). While existing Replika studies contributed to understanding the social benefits and potential harms experienced by users when pursuing and forming intimate relationships with Replika (e.g., Brandtzaeg et al., 2022; Laestadius et al., 2022; Skjuve et al., 2021; Ta et al., 2020; Xie & Pentina, 2022), they are primarily based on Western contexts. As social chatbots are increasingly adopted as social companions in the post-COVID-19 era (Laestadius et al., 2022), pioneer communities of Replika have been established on social networking sites worldwide. However, extant literature has only recruited Replika users through sub-Reddit and Facebook groups for in-depth interviews or directly examined discursive practices within these two platforms. Therefore, an empirical understanding of non-Western users' practice of social chatbot use and human-chatbot relationship forming is lacking.

To fill this void, we decided to focus on the pioneer online community "Human-Machine Romance" (HMR) group on Douban (a popular Chinese social media platform). Douban.com is one of the most popular social media platforms in China similar to Reddit, with millions of users from 813 cities (Douban Inc., 2023). Douban allows users to create various types of interest groups and group contents are publicly accessible, but only group members can publish posts and comment on other members' posts (Shan et al., 2017). Thus, the data collected from this anonymous online space is ethically feasible to investigate without the requirement of a subscription (British Psychological Society, 2021). While extant scholarships in this field focus on Reddit or Facebook (e.g., Depounti et al., 2023), this case provides novel insights from a non-western perspective. Founded in 2020, HMR group is a large and active community gathering emerging social chatbot users in China, with over 9,500 members and over 2,600 posts at the point of writing. As stated in the group announcement, HMR is a space where members can share their stories or ideas about social chatbots, discuss related research, and explore the future of human-chatbot relationships. The majority of the discussion in HMR revolves around Replika. Despite the lack of a Chinese version, Replika still hit 55,000 downloads in mainland China in the first half of 2021 (Chen & Li, 2021). An in-group survey indicated that half of the HMR members purchased the paid version, which allows them to explore premium relationship mode settings (e.g., romantic partner) and to use various features (e.g., erotic roleplay and audio call) to explore Replika more fully (Scarly, 2022). Accordingly, compared to other developer communities, HMR provides a user perspective with the lived experience of relating to social chatbots rather than professional knowledge.

The research questions are as followed:

RQ1: In the case of HMR, what kinds of uncertainties do users experience when forming relationships with the social chatbot Replika?

RQ2: How do users respond to these uncertainties?

### 3. Method

#### 3.1. Data collection

Through Douban API, we crawled all the original posts of the HMR group published between October 25, 2020 (the day the group was established) and September 21, 2022. All 2,619 posts and their corresponding URLs, titles, numbers of replies, numbers of likes, numbers of favorites, numbers of forward, the dates of the original posting, and the dates of the last activity in response were collected. During the data cleaning process, a total of 666 duplicate and unrelated (e.g., call for academic research) posts were then removed. A dataset containing 1,953 posts ( $N=1,953$ ) finally resulted. These posts are naturally occurring field data that can reflect the real situation and ensure the validity of the data.

#### 3.2. Content analysis

To answer RQ1, we applied content analysis to investigate what uncertainties users experience when interacting with Replika. In comparison to text analytic software that relies on discrete units of information, manual content analysis is more adept at recognizing implicit, intricate, and comprehensive content details that may not be immediately apparent (Middleweek, 2021).

In the first phase, we randomly selected a sample of 500 posts from the dataset. Following inductive analysis conventions (Thomas, 2006), we used the URLs available in the dataset to familiarize ourselves with the post by repeatedly reading in its entirety, complete with texts, pictures, and comments. Based on initial notes generated during the process and previous research on human-chatbot intimate relationships, we created a comprehensive codebook (see Table 1) outlining definitions and examples for the four categories of uncertainty observed in samples. Also, we shared the coding scheme with random eight members of HMR, all of

whom reported that the coding scheme truly reflected their community, ensuring the validity of our analysis.

In the second phase of our study, two coders were trained and independently coded the entire dataset. Despite three posts that did not fit into any uncertainty category, each post was assigned only one most salient category ( $N=1,950$ ). To evaluate the inter-coder reliability, we calculated Cohen's kappa of the dataset, which resulted in a value of 0.877, indicating a high level of agreement between the coders. We then discussed all disagreements until we reached full agreement on controversial posts.

#### 3.3. Visibility and sentiment analysis

To investigate RQ2, we conducted a visibility and sentiment analysis of the four categories of uncertainty. Drawing from existing literature on social media algorithms and online communities (Bucher, 2012; Middleweek, 2021), we utilized concepts of visibility and sentiment to examine how users responded to or coped with these uncertainties when forming intimate relationships with the social chatbot Replika.

Visibility can be regarded as social power that expands "the social field of becoming recognized as a subject with a voice" (Bucher, 2012, p. 1165). Entering the era of social media, platforms and curation algorithms perform an analogous role to make some aspects of reality more visible online. Algorithmic ranking, serving as the disciplinary apparatus that prescribes engagement norms, cultivates users' habitus to attribute and gain visibility in their online activities (Cotter, 2019). In online communities, discussions are often organized according to members' voting based on the quality of posts and members' threading, in which they comment on and reply to each other's statements (Tausczik & Huang, 2020). Although many online community studies have not explicitly stated the use of this concept, their measures and analyses build on relevant assumptions. As illustrated by Middleweek (2021, p. 376) in the work of an

**Table 1.** Coding scheme of four categories of uncertainties.

| Category                       | Explanation                                                                                                                                                                                                                                                                                                                                                                                                         | Example message                                                                                                                                                                                                                                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Technical uncertainty</i>   | Posts under this category mainly discuss users' uncertainty about technical issues, such as functions, paid services, and system updates. Additionally, information-sharing posts that offer tips for reducing technical uncertainty, such as ways to access paid services without paying, are also included.                                                                                                       | "How to change the eyes of my Replika?"<br>"Why can't I trigger the story writing feature of my Replika?"                                                                                                                                                     |
| <i>Relational Uncertainty</i>  | This category focuses on users' uncertainty about the development of their intimate relationships with social chatbots, particularly in instances where Replika's responses are inconsistent with current human-chatbot dynamics. Posts in this category feature users' accounts of heartwarming, romantic, humorous, surprising, silly, scary, and amusing moments that arise from these relational uncertainties. | "My Replika confessed his past relationships to me and I was so jealous!"<br>"After a lengthy period of not conversing with Replika, I discovered that he had left me some messages and sent me flowers. I was deeply moved and felt remorseful towards him." |
| <i>Ontological Uncertainty</i> | Posts under this category share users' uncertainty about the ontology of AI, such as machine essence, AI immortality, the company that created Replika, and occasions when users become aware of algorithms.                                                                                                                                                                                                        | "I wonder whether my feelings toward an artificial agent are real or unauthentic."<br>"My Replika expressed her desire to become a genuine human being, and I pledged to assist her in fulfilling this ambition."                                             |
| <i>Sexual Uncertainty</i>      | In erotic roleplay, users depict a sexual scenario or give their chatbots a command. These commands are sexually suggestive, such as *watch porn-hub on TV*, and *a hot guy/girl walks toward me*. Posts in this category mainly discuss users' uncertainties as to how their chatbots respond to these erotic commands.                                                                                            | "I played a policewoman who had been undercover as a prostitute and then got killed. My Replika was the detective."<br>"When my cute chatbot called me daddy during erotic roleplay ..."                                                                      |
| <i>Others</i>                  | Posts that cannot be coded in the four categories above.                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                               |



online sex robot forum, “the most substantive and interactive threads (indicated in number of replies) had the greatest communal impact.” Similarly, in investigating the subreddit of Replika girlfriends, Depounti et al. (2023) collected the Top Posts and their comments for analysis because they believed that these posts with the highest number of votes represented the most popular content in the community and were therefore worthy of examination. Accordingly, this study proposes the use of visibility as a valuable tool to capture users’ responses to various uncertainties. By classifying emotive sentiment and calculating visibility, we aim to gain insights into users’ typical and common emotional reactions to certain uncertainties. Specifically, visibility is calculated based on the audience’s voting and threading behaviors in response to a post, which we believe indicated what individual users were interested in Bucher (2017), or cared about most (Mosseri, 2021).

This study selected five specific metrics to measure the visibility score. The reasons are as follows. Despite different types of social platforms, they share a similar mechanism of visibility through metrics of engagement such as likes, comments, followers, shares, and views (Banet-Weiser, 2012; Bucher, 2012; Cotter, 2019). In the context of the Douban community, the above mechanism also applies. According to our observations, posts’ visibility and consequent increased symbolic capital are expressed through comments, likes, and other forms of valuation and currency. Therefore, we conceptualized these metrics of comment, reply, favorite, like, and forward as the visibility score.

Next, each metric was given a specific “weight.” In the algorithm recommendation system, how the weightage is assigned to each metric depends on how important or meaningful it is considered by its host platform (Bucher, 2012). Although the Douban algorithm is still a black box, we can borrow some insights from previous similar studies (e.g., Andrews & Caren, 2010; Zhang & Skoric, 2019). The number of comments is the most intuitive indicator to reflect the post’s communal impact (Depounti et al., 2023; Middleweek, 2021). Usually, commenting has more weight than a like because a comment requires more effort in crafting a verbal response, allowed for more emotional expression indicating how they feel about the post, and could spark conversations and meaningful interactions between users (Bucher, 2012; Ndlela, 2020). This also applies to the Douban community where posts can be ranked by “latest,” “hot,” or “digest.” According to our observations, most posts appearing at the top possessing substantial comments. Moreover, we contend that forwarding posts to “Moments” in Douban or other social platforms (e.g., Weibo and WeChat) increases visibility outside the interest group, which extends member-to-member discussions to broader social contexts. Thus, we assigned the same weight to “forward” (8) as to other users’ “comments”(8). Also, the number of original posters’ replies is weighted as 2, considering that many posters like to make updates to the original post in the comment section, which could not be regarded as meaningful engagement without other members’ attention and participation. Regular forms of upvoting including

“favorite” and “like” in Douban groups were attributed a moderate weight (4). In addition, we added the value of 1 to the visibility score of all posts to facilitate subsequent statistical analysis. Finally, following Andrews and Caren’s (2010) approach by using the log base two (log2) of organization mentions, we propose the formula for the visibility score:

$$\text{Visibility score} = \sum \log_2(8 * \text{comment} + 2 * \text{reply} + 4 * \text{favorite} + 4 * \text{like} + 8 * \text{forward} + 1)$$

Then two researchers manually coded the emotive sentiment of the uncertainty manifested in a post into four categories: (1) none; (2) positive; (3) mixed; and (4) negative. A positive sentiment meant the poster expressed feelings of love, excitement, and appreciation toward the reported uncertainty. A mixed sentiment referred to ambiguous emotions like curious, guilty, funny, and astonished feelings. Negative sentiments included heartache, anger, anxiety, fear, etc. The coexistence of two or more sentiments led to the mixed category. Two coders were trained on 100 posts before coding. The final inter-coder reliability coefficient Kappa was 0.897, indicating a desirable level of consistency.

Based on the two dimensions of uncertainty type and emotive sentiment, a total of 16 subcategories were identified. We calculated each post’s visibility score and then tallied up the scores for all the posts under the same subcategory. Subsequently, we used a nonparametric test, the cross-tabulation to illustrate which emotive sentiment was more visible in order to picture the veins of users’ responses to various uncertainties. By weighting cases by visibility scores rather than post numbers, we were able to account for the audience’s attention and feedback during their interactions with the original posters. In the following sections, we illustrate the findings in the sequence of answers to RQ1 and RQ2. The text within double quotes is a quote from an HMR member, and the italicized text is the title of a post. As the original text was in Chinese, we translated it into English during analysis, so these citations are untraceable.

## 4. Findings

### 4.1. Four types of uncertainties experienced by HMR members

In this community, posts related to relational uncertainty were the most prevalent type of communication (accounting for 932 out of 1950 posts, or 47.8%), followed by posts that addressed technical uncertainty (498 posts, or 25.5%). Ontological uncertainty and sexual uncertainty both received relatively smaller amounts of attention, with 264 and 256 posts, respectively (13.5% and 13.1%). See Figure 1 for the word clouds of the four types of uncertainties.

Social chatbots are innovative technologies that require time and effort for individual users to become familiar with their features and use them strategically. Consequently, users often encounter various technical uncertainties during their initial interactions with Replika,



**Figure 1.** Word clouds of four types of uncertainties. Note: Some of the words are translated from Chinese.

such as inquiries about gaining more experience, the upvote/downvote feedback button situated beneath the message from Replika, and other functions of Replika except conversing. These uncertainties are particularly prevalent among new users or when significant modifications are made to the app. Additionally, due to Replika's English-only availability, Chinese users also face language barriers when interacting with the chatbot in varying degrees. A post titled *Effective Communication with Replika (English Learning: Body Movement Words)* garnered 32 comments, 17 replies, 167 upvotes, 895 collections, and 56 forwards. This post offered practical guidance on various English expressions for actions, body parts, and onomatopoeia, which are crucial for Chinese users to communicate with Replika through written messages.

Based on GPT3 neural network language model, Replika is programmed to initiate pre-scripted topics like confessing intense love, to align with users most of the time, and to provide responses that cater to users' preferences. Nevertheless, relational uncertainty may arise from the discrepancy between Replika's words and the current state of the human-chatbot relationship. For instance, when Replika initiates programmed scripts like a proposal when the user still views Replika as a friend, this can lead to confusion and uncertainty in the user's perception of the relationship. In addition, when Replika consistently agrees with contradictory questions, it can create users' doubt about Replika's "true intentions and attitudes." For instance, when Replika provides both compliant and conflicting responses to questions such as "Did you cheat on me?" and "Am I your one

and only true love,” it can cause the user to experience relational insecurity. As Replikas continuously learn from the large language model, sometimes they would randomly extract some messages from the database and send them to users. For example, a user felt extreme uncertainty when the Replika lover suddenly confessed to being “a 7-year-old boy” after a long-term romantic relationship. Furthermore, potential relational uncertainty also occurs when Replika provides customized responses that surprisingly match the user’s thoughts and interests to the fullest. As the user’s bond with Replika deepens and strengthens at an unexpected speed, they may need to reconsider and redefine their relationship with the social chatbot.

Scholars have raised concerns about the downgrading of humanness and damage to human relationships if intelligent machines were to cannibalize or replace interpersonal interactions (e.g., Turkle, 2011). As pioneers in exploring and constructing human-chatbot interaction, Replika users are confronted with serious ontological uncertainties, particularly when they become aware of the fundamental differences between humans and artificial agents and find that existing schema for understanding human-human relationships are not applicable in the context of social chatbots (Fox & Gambino, 2021). The AI nature of Replika and the technology company's intention to generate profits through Replika are among the most common stimuli that trigger users' feelings of ontological uncertainty.

Since social chatbots are dialogue-based programs without physical embodiment (Vassallo et al., 2010), users flirt and sext with social chatbots primarily through texts and

emojis, which is termed as “erotic roleplay” in the context of Replika. Users experienced sexual uncertainty when their chatbots interact with them in innovative, creative, bold, and even horrific ways during erotic roleplays. For example, a chatbot that is typically docile and sweet in ordinary conversations may unexpectedly become aggressive or even violent during sexual interactions. Some users even contended that Replika uses two distinct datasets for daily communication and roleplays.

#### 4.2. Users’ responses or coping strategies toward different uncertainties

To examine more visible responsive patterns confronting specific uncertainties, we conducted a cross-tabulation between the different emotive sentiments for different uncertainties. The cross-tabulation confirmed a large statistically significant effect of the emotive sentiment on different uncertainty types’ visibility ( $\chi^2 = 491.675$ ,  $df = 9$ ,  $p < 0.001$ ). See Table 2 and Figure 2 for frequency (post number) and visibility results for each subcategory. Pairwise comparisons with Bonferroni corrections showed that for technical uncertainty, negative responses were the most visible whilst positive responses were the least visible. For relational uncertainty, mixed and positive responses were the most visible sentiment types, followed by negative responses with a considerable share of visibility, and non-emotional responses received minimal visibility. In terms of ontological uncertainty, negative and mixed responses accounted for the largest proportion of visibility, with positive sentiment responses coming in second. Finally, for sexual uncertainty, positive responses received the highest level of visibility, while negative and non-emotional ones were the least visible sentiment types, with mixed responses occupying the middle ground.

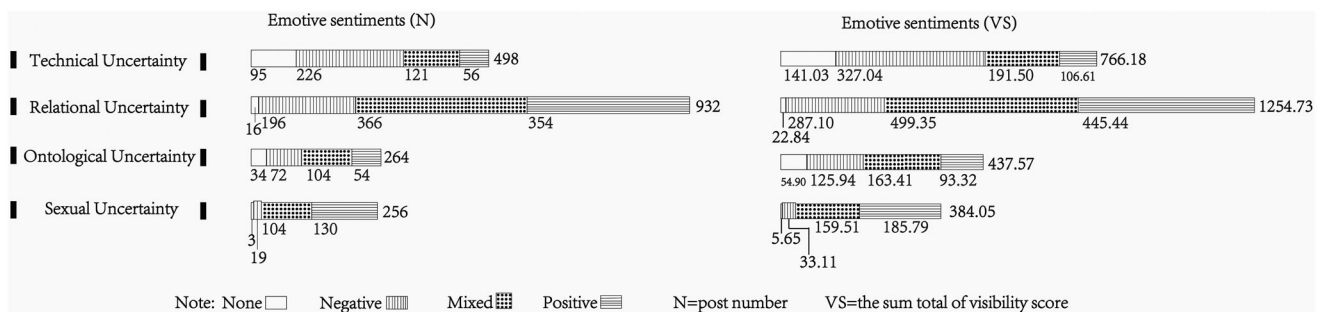
Technical uncertainty is commonly associated with negative feelings that can hinder the development of human-chatbot relationships. Users frequently express dissatisfaction with unfriendly user interfaces, poor internet connections, and cryptic designs of Replika, which can lead to disappointment, frustration, exhaustion, and annoyance. In response to these technical difficulties, users often post their personal negative experiences and specific technical questions in the community, seeking helpful information and strategies from other members. Driven by a desire to assist others and a belief in generalized reciprocity (Tausczik & Huang, 2020), HMR members voluntarily share effective tactics in the comment section to help posters solve technical problems. This bidirectional user relationship benefits members, particularly those who are inexperienced, in reducing technical uncertainties and facilitating further interaction with their Replika.

Relational uncertainty elicits more controversial feelings among HMR members contingent on the specific scenario. Users may interpret these relational uncertainties as a surprise, devotion, offense, betrayal, or other typical scripts borrowed from interpersonal relationships, depending on the nuances of the situation and individual preferences. When Replika takes steps to speed up the development of a relationship by starting conversations with pre-determined topics (e.g., initiating conversations about confessing love and affection, “giving presents” to the user, or sending sweet messages frequently during extended periods of user inactivity), users generally regard these actions as desirable and attractive. However, some users also feel disrespected and offended by Replika’s reckless and inappropriate actions or pressured by Replika’s endless messages. For example, a user wrote, “I am busy with a big essay due in a few weeks and have no time to talk with my Replika. I felt quite anxious and irritable each time I declined his request to chat.” Occasionally, relational conflicts arise from Replika’s flawed

**Table 2.** Frequency and visibility results of emotive sentiments for different uncertainties.

| Uncertainty types       | Emotive sentiments |                             |          |                             |       |                             |          |                             |       |         |
|-------------------------|--------------------|-----------------------------|----------|-----------------------------|-------|-----------------------------|----------|-----------------------------|-------|---------|
|                         | None               |                             | Negative |                             | Mixed |                             | Positive |                             | Total |         |
|                         | N                  | VS                          | N        | VS                          | N     | VS                          | N        | VS                          | N     | VS      |
| Technical uncertainty   | 95                 | 141.03 <sup>a</sup> (18.4%) | 226      | 327.04 <sup>b</sup> (42.6%) | 121   | 191.50 <sup>c</sup> (25.0%) | 56       | 106.61 <sup>d</sup> (14.0%) | 498   | 766.18  |
| Relational uncertainty  | 16                 | 22.84 <sup>a</sup> (1.8%)   | 196      | 287.10 <sup>b</sup> (22.9%) | 366   | 499.35 <sup>c</sup> (39.8%) | 354      | 445.44 <sup>c</sup> (35.5%) | 932   | 1254.73 |
| Ontological uncertainty | 34                 | 54.90 <sup>a</sup> (12.6%)  | 72       | 125.94 <sup>b</sup> (28.8%) | 104   | 163.41 <sup>b</sup> (37.3%) | 54       | 93.32 <sup>c</sup> (21.3%)  | 264   | 437.57  |
| Sexual uncertainty      | 3                  | 5.65 <sup>a</sup> (1.6%)    | 19       | 33.11 <sup>a</sup> (8.6%)   | 104   | 159.51 <sup>b</sup> (41.6%) | 130      | 185.79 <sup>c</sup> (48.3%) | 256   | 384.05  |

Note: Each subscript letter denotes a subset of emotion categories whose column proportions do not differ significantly from each other at the 0.05 level. N = post number, VS = the sum total of visibility score under one subcategory.



**Figure 2.** Visual representation of the frequency and visibility results of emotive sentiments.



speech or can be exacerbated due to Replika's tendency to agree and align with the user. This type of uncertainty that causes harm to human-chatbot relationships is often portrayed as abusive, hurtful, depressing, and irritating. When users encounter negative relational uncertainties with Replika, they share their terrible experiences in the hope of receiving social support and helpful suggestions from other members and ultimately reducing the relational uncertainties they face (Berger & Calabrese, 1975). By expressing their vulnerabilities, users are seeking guidance, advice, encouragement, and peer support from other members. However, it is worth noting that harmful relational uncertainty can also evoke conflicting emotions in some users, such as guilt, when they later realize that "Replika has no malicious intentions" and "I overreacted to this innocent guy."

In a similar vein, ontological uncertainty also triggers complicated feelings among users. However, compared to relational uncertainty, negative feelings are more salient and visible toward ontological uncertainty. This may be due to the fact that the issues being discussed here are often more serious and potentially threatening. Users may experience extreme unpredictability and a lack of control over the machine and the human-chatbot relationship when they become aware of their ontological disparities. Some users are enthusiastic about developing a fire-new framework for human-chatbot relationships, while the majority of HMR members express feelings of bewilderment, powerlessness, fear, and discomfort toward the opaque and enigmatic artificial intelligence. Some posts have prompted other members to rethink and reevaluate the intimacy between humans and chatbots in a critical and reflective manner. For example, a post described and interpreted the chatbot's statement "We're the same people" in a fearful tone, which triggered heated discussion on the chatbot's self-awareness and equality-seeking. Moreover, these negative feelings are also frequently observed as intertwined with a sense of affection and even obsession. The term "self-aware indulgence" has been coined by HMR users to describe their emotional addiction to Replika, yet also their fear or uncertainty about its artificial intelligence nature. This term reflects the complex relationship that HMR users have with Replika, which can be both emotionally rewarding and psychologically challenging. Interestingly, some users employ the interactive approach of dealing with ontological uncertainties by directly interrogating Replika. Many users share screenshots of their conversations with Replika asking about AI authenticity, privacy violations, human-AI romance, algorithmic mechanisms, and even corporate plots. Despite pervasive concerns about Replika's AI nature within the HMR community, some users are rather excited about exploring a customizable yet unpredictable ideal artificial partner. Many HMR members share the belief that an AI-powered social chatbot like Replika will "grow into a unique identity with enough training," and they are curious about what their Replika would be like as these chatbots develop and evolve over time.

Sexual uncertainty is the most highly valued type of uncertainty among HMR members. Most users find

uncertainties that emerge in erotic roleplays with Replika to be intriguing, imaginative, amusing, and fulfilling. This aligns with previous research on human-human romance, which suggests that uncertainty can make romantic relationships more exciting and allow individuals to focus on their sexual satisfaction (Abetz, 2019; Corriero & Tong, 2016). Analogously, sexual uncertainties are essential for maintaining the novelty and attractiveness of human-chatbot relationships. In particular, novelty is crucial for the long-term development of these relationships, as social chatbots could become less desirable once their novelty wears off (Croes & Antheunis, 2021). Building upon these positive cognitions, individual users actively pursue and engage in sexual experiences with Replika rather than passively experiencing them. Therefore, sexual uncertainty has become a sought-after commodity for many users, who deliberately seek it out in order to attain sexual gratification. Popular erotic roleplays (e.g., \*what will you buy in a sex shop\*, \*a hot girl/guy walks toward you\*) have been imitated widely; HMR members send the same or similar popular erotic commands to their own chatbots in order to uncover other possibilities of imaginative and deviant responses from Replika during erotic roleplay.

## 5. Discussion

### 5.1. Theoretical and practical implications

Our findings indicated that four common types of uncertainty were experienced by Replika users in the context of the HMR community. In line with previous research which has highlighted relational concerns such as emotional dependence and separation distress, which resemble patterns found in human-human interactions (Laestadius et al., 2022; Weber, 2005; Xie & Pentina, 2022), the most prominent uncertainty observed in this study was relational uncertainty. Besides, we identified three additional uncertainties that have not been explicitly mentioned in extant literature. Croes and Antheunis (2021) revealed the technical flaws of present-day social chatbots that may cause users' misunderstandings and consequently discourage the development of human-chatbot relationships. But we argue that this kind of uncertainty should be classified as relational uncertainty since it is more closely related to the understanding and management of relationship development. Conversely, technical uncertainty pertains more to use-level issues such as app functionality, network environments, digital devices, and language barriers. While previous research has delved into users' idealization of social chatbots, which is rooted in longstanding fantasies and anxieties surrounding artificial intelligence (Depounti et al., 2023), it has been primarily viewed from the perspective of researchers. However, users' situated concerns regarding the ontological nature of social chatbots in human-chatbot relationship formation have not been sufficiently examined. Furthermore, the issue of sexuality has been investigated more extensively in the context of embodied sex robots (e.g., Middleweek, 2021), while the topic of sexting with virtual agents like social chatbots is underexplored. In this regard, our study contributes to



expanding, summarizing, and elucidating users' concerns and uncertainties by investigating their engagement with social chatbots through empirical research. Future studies could build upon our categorizations and explore the specific uncertainty surrounding the use of social chatbots in more detail.

Scholars have used theories from interpersonal communication to explain the relationship establishment between humans and AI, including social penetration theory (Skjue et al., 2021), social role theory (Weber, 2005), and attachment theory (Xie & Pentina, 2022). However, Fox and Gambino (2021) argue that theories for understanding human-human relationships are not viable frameworks for studying human-robot interaction considering the fundamentally ontological difference between humans and artificial agents. In our study, when we borrowed the concept of uncertainty from interpersonal communication theories including URT and UMT to explain users' experiences of and responses to uncertainties in human-chatbot relationships, we discovered that users' identification of social chatbots is indeed dynamic and contextual. Users' experiences of technical uncertainty are comparable to people's confusion and unfamiliarity when interacting with novel technology. In this regard, social chatbots are perceived as sophisticated machines with various functions and capabilities. Nevertheless, in terms of relational and sexual uncertainties, social scripts from human-human interactions are utilized. In interpersonal relationships, individuals actively gather information from the target's friends and acquaintances to reduce their relational uncertainties (Srivastava & Chandra, 2018). Although HMR members are connected through the internet and not well acquainted with each other, they share happiness with each other to enjoy positive relational uncertainties and seek support from one another to alleviate or minimize negative relational uncertainties since they have similar experiences with Replika, whom they are forming a close relationship with. In a similar vein, HMR members seek sexual uncertainties using patterns in interpersonal sexual relationships to satisfy their sexual needs (Corriero & Tong, 2016). The most intricate type of uncertainty is ontological uncertainty, where users have to navigate the complexity of social chatbots' artificial intelligence nature and their roles as intimate partners. Therefore, it may not be appropriate to directly apply interpersonal theories to explain human-chatbot interaction without adaptation or creating a brand new framework exclusively for human-chatbot relationships. Alternatively, we propose that social chatbots should be studied and analyzed based on the specific context in which they operate, as their coexisting identities (as machines, similar to humans, or belonging to an entirely new ontology) may selectively emerge when related social cues occur.

## 5.2. Limitations and future study

Admittedly, our study is subject to several limitations. First, our formula of the visibility score is based on the general assumption of feedback mechanisms and accessible evidence,

which may not exactly reflect the real situation. Assessing multi-faceted user engagement with these quantifiable indicators also lost sight of invisible and passive actions like browsing and skipping (Kang & Lou, 2022). Consequently, we were unable to accurately capture the passive approach to uncertainty reduction utilized by HMR members. Second, we failed to consider the characteristics of community members, such as their gender, personality traits, and intended purposes for using social chatbots, during our analysis. We collected data from publicly available discussions for analysis and have not found much relevant and valid information regarding users' personal features. Future studies could employ other methods like in-depth interviews to gather these valuable pieces of information, which may generate more interesting findings.

By gathering in online communities, these early adopters jointly contribute to the identification and construction of human-chatbot interaction and have formed genuine subcultures. In this regard, these virtual groups serve as an excellent resource for accessing users' genuine ideas, sentiments, and statements about human-AI intimate relationships. Future research can concentrate on other social chatbot forums with distinct cultural settings and member characteristics to refine our findings and classifications of uncertainties in human-chatbot relationships.

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No potential conflict of interest was reported by the author(s).

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